Supplementary material – ESM 2 – Detailed selection criteria

ABSTRACT

Inclusion criteria

Study population:

- Population: people with diabetes infected with COVID-19
  - People with diabetes diagnosed at any point/ had any duration of time with diabetes.
  - any age
  - any gender
  - human

Intervention to the population:

- none (purely observational)

Clinical outcome:

- Poor COVID-19 prognosis
  - Outcome of poor prognosis is defined as any COVID-19 related poor outcome such as: critical care, mortality, hospital admission, intubation, mechanical ventilation, increased hospital stay, sepsis, pneumonia, artificial respiration, ICU admittance, and any other COVID-19 related adverse complication

Study design

- **MUST BE: Non-interventional observational prognostic studies**: these are observational studies where main measurement is trying to identify whether certain prognostic factors/ prognostic tool/ risk factors/ independent risk factors/ predictors/ phenotypes/ clinical, biological or laboratory characteristics/ clinical features/ patient history factors are associated with poorer COVID-19 prognosis (as already defined) *within the population with diabetes* **NOT** between population with and without diabetes. This may include the development of a risk score, prediction/risk model, prognostic tool.

  Phrases in the literature to look out for might be similar to:

  - 'aim to describe the clinical characteristics and outcome of hospitalised covid-19 patients with diabetes'
  - 'aim to develop risk stratification tool'

        - ‘prognostic tools/ risk stratification tool/clinical score to help stratify/predict the COVID-19 infected population with diabetes’
        - ‘assess the prognostic value/association of X features of diabetes and X poor COVID-19 outcome’
        - ‘clinical features, and outcomes were compared between DM patients admitted to the intensive care unit (ICU) or not’
        - ‘compare diabetic phenotypic characteristic between having X poor COVID-19 outcome and not having X poor COVID-19 outcome’
        - ‘explore the risk factors for in-hospital death and complications of coronavirus disease 2019 (COVID-19) patients with diabetes’
        - ‘investigate risk factors for and build a cross-validated predictive model of fatal or critical care unit-treated COVID-19 among people with diabetes’
        - ‘investigate risk indicators of in-hospital mortality and severity of coronavirus disease-2019 (COVID-19) in patients with diabetes mellitus (DM)’
        - ‘Covid-19 fatality prediction in people with diabetes’
    - **target group:** People with diabetes with a certain potential prognostic factor/ risk factor as described above.
        - preferably this should include pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement) as a potential prognostic factor/ risk factor however it is not essential for the abstract includes this.
    - **comparison group:** Preferably comparison is *within* the population with diabetes itself, for example to people with diabetes without the respective potential prognostic factor/ risk factor etc to act as a control. However, it is not essential for the abstract to define a comparison at this stage.
    - other than non-interventional or prognostic, observational studies can be identified as cohort studies or cross-sectional studies.
    - these studies can be monocentric or multicentred
    - these studies can be retrospective or prospective
    - these studies MUST include poor COVID-19 prognosis as a COVID-19 outcome.

- these studies MUST report measurement with regards to potential prognostic factors for worse COVID-19 outcomes in the population with diabetes.

Sample size

- any

Setting

- hospital/ICU/medical

Data type:

- primary data
- quantitative
- all studies must report relevant outcome data, however at the abstract stage the way this data is reported is not essential.

Language:

- any language

Geographical location:

- global

Time period:

- COVID-19 pandemic (Dec 2019 – current)

Publication date restriction

- December 2019- June 2021 (within COVID-19 pandemic)

<u>Exclusion Criteria:</u>

Study population:

1. People with diabetes not infected with COVID-19
2. not relating to COVID-19
3. People without diabetes
4. animal studies

Intervention:

- any

study type:

- ANY paper that is NOT a **non-interventional observational prognostic study,** such as:

- any non-observational study: RCTs, reviews, editorials.
- interventional observational studies
- observational study where main measurement is NOT trying to identify prognostic factors/ risk factors/ risk scores/ predictors/ prediction models/ phenotypes/ clinical, biological or laboratory characteristics/ patient history factors that are associated with poor COVID-19 prognosis (as already defined) *within the population with diabetes.*
- observational study without a comparison: case-reports
- observational study simply assessing diabetes mellitus to be a risk factor for poorer COVID-19 prognosis, not risk factors for worse prognosis within the population with diabetes.

Clinically relevant outcome:

- Did not measure poor COVID-19 prognosis as a main COVID-19 outcome.

Data type:

- no data reported relevant to measured outcome with regards to potential prognostic/risk factors etc
- data collection ongoing

Time period:

- Outside of the COVID-19 pandemic (Dec 2019 – current)

Publication date restriction

- Outside of 2019 – June 2021

<u>WHAT TO DO FOR PAPERS THAT DO NOT HAVE ABSTRACTS:</u>

- Apply abstract exclusion criteria to the title
- where title indicates a potentially relevant study/ where can't apply criteria to the title/ unclear the proceed to full paper screening and application of relevant criteria

<u>Full paper screening</u>

<u>Inclusion Criteria</u>

Study population:

- Population: people with diabetes infected with COVID-19

- People with diabetes diagnosed at any point/ had any duration of time with diabetes.
- any age
- any gender
- human

Intervention to the population:

- none (purely observational)

Index prognostic factor:

- microvascular disease ((including retinopathy within that measurement or as a separate measurement)

Clinical outcome:

- Poor COVID-19 prognosis
  - Outcome of poor prognosis is defined as any COVID-19 related poor outcome such as: critical care, mortality, hospital admission, intubation, mechanical ventilation, increased hospital stay, sepsis, pneumonia, artificial respiration, ICU admittance, and any other COVID-19 related adverse complication

Study design

- **MUST BE: Non-interventional observational prognostic studies**: observational studies where main measurement is trying to identify prognostic factors/ risk factors/ risk scores/ independent risk factors/ predictors/ prediction models/ phenotypes/ clinical, biological or laboratory characteristics/ patient history factors that are associated with poor COVID-19 prognosis (as already defined) *within the population with diabetes.*
  - **target group:** People with diabetes with a certain potential prognostic factor/ risk factor etc as described above.
    - of the prognostic/risk factors etc measured, this **MUST** include measurement of people with diabetes with pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement)
      - pre-existing microvascular disease may be defined in-text as microvascular disease with a series of measurements of which

MUST include retinopathy, retinopathy as an independent measurement, 'visual complications of diabetes' which MUST include retinopathy, microangiopathy (which MUST include retinopathy). Overall all of these measurements MUST include a measurement for retinopathy.

- **comparison group: MUST** have a comparison group.
  - comparison is **within** the population with diabetes itself, for example to people with diabetes without the respective potential prognostic factor/ risk factor **OR** people with diabetes with respective potential prognostic factor but *not* with the measured COVID-19 outcome.
- these studies can be monocentric or multicentred
- these studies can be retrospective or prospective
- these studies MUST include poor COVID-19 prognosis as their main COVID-19 outcome.
- these studies MUST measure potential prognostic factors for worse COVID-19 outcomes within the population with diabetes.
- these studies MUST include pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement) as a potential prognostic factor for worse COVID-19 outcomes within the population with diabetes.

Sample size

- any

Setting

- hospital/ICU/medical

Data type:

- primary data
- quantitative
- all studies must report relevant outcome data in the form of either comparative sample numbers or prognostic effect estimates such as: odds ratio (OR), relative risk (RR)

- this MUST include data with regards to pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement) as a potential prognostic/risk factor for poor COVID-19 prognosis (as already defined) *within* the *population with diabetes*.

Language:

- any language

Geographical location:

- global

Time period:

- COVID-19 pandemic (Dec 2019 – Current)

Publication date restriction

- December 2019- June 2021 (within COVID-19 pandemic)

<u>Exclusion Criteria</u>

Study population:

5. People with diabetes not infected with COVID-19
6. not relating to COVID-19
7. People without diabetes
8. animal studies

Intervention:

- any

study design:

- ANY paper that is NOT a **non-interventional observational prognostic study, such as:**
    - any non-observational study: RCTs, reviews, editorials.
    - interventional observational studies
    - observational study where main measurement is NOT trying to identify prognostic factors/ risk factors/ risk scores/ predictors/ prediction models/ phenotypes/ clinical, biological or laboratory characteristics/ patient history

factors that are associated with poor COVID-19 prognosis (as already defined) *within* the *population with diabetes*.

    - observational study without a comparison: case-reports
- **Non-interventional observational prognostic study** that does NOT include pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement) as one of the measured potential prognostic/risk factor for poor COVID-19 prognosis (as already defined) *within* the population with diabetes.
- **Comparison:** outside of the population with diabetes ie people without diabetes.

Clinically relevant outcome:

- Did not measure poor COVID-19 prognosis as the main COVID-19 outcome.

Data type:

- no data with regards to prognostic effect estimates/ comparative sample numbers
- no data with regards to pre-existing microvascular disease (including retinopathy within that measurement or as a separate measurement) as a potential prognostic/risk factor for poor COVID-19 prognosis (as already defined) *within the population with diabetes*.
- data collection ongoing

Time period:

- Outside of the COVID-19 pandemic (Dec 2019 – Current)

Publication date restriction

- Outside of December 2019 - June 2021